

# Vulnerability Assessment Report

1<sup>st</sup> January 2025

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## System Description

The server hardware consists of a powerful CPU processor and 128GB of memory. It runs on the latest version of Linux operating system and hosts a MySQL database management system. It is configured with a stable network connection using IPv4 addresses and interacts with other servers on the network. Security measures include SSL/TLS encrypted connections.

## Scope

The scope of this vulnerability assessment relates to the current access controls of the system. The assessment will cover a period of three months, from June 2025 to August 2025. [NIST SP 800-30 Rev. 1](#) is used to guide the risk analysis of the information system.

## Purpose

The database server is a critical asset to the business as it stores and manages essential data, including employee records and login activity, which are crucial for daily operations and security monitoring. Securing the data on this server is vital to protect sensitive information, prevent unauthorized access, and maintain customer and employee trust. If the server were disabled or compromised, it could disrupt operations, lead to data breaches, financial losses, and damage the organization’s reputation. This vulnerability analysis aims to identify and mitigate potential risks to ensure data integrity, confidentiality, and business continuity

## Risk Assessment

| Threat source | Threat event                                                                                    | Likelihood | Severity | Risk |
|---------------|-------------------------------------------------------------------------------------------------|------------|----------|------|
| Competitor    | Obtain sensitive information via exfiltration                                                   | 1          | 3        | 3    |
| Hacker / APT  | <b>Conduct Denial of Service (DoS) attacks</b> – Send excessive automated requests to overwhelm | 2          | 3        | 6    |

|                                    |                                                                                                                  |   |   |   |
|------------------------------------|------------------------------------------------------------------------------------------------------------------|---|---|---|
|                                    | <i>database operation</i>                                                                                        |   |   |   |
| <i>Hardware / Software Failure</i> | <b>Disrupt mission-critical operations</b> – Database unavailability due to disk failure or hardware malfunction | 2 | 2 | 4 |

## Approach

The selected threat sources and events were chosen based on their potential impact on business operations and data security. A **competitor attempting to exfiltrate sensitive information** poses a significant business risk due to the potential exposure of confidential data, which could harm competitive advantage, even though the likelihood is low. **Hackers or Advanced Persistent Threats (APT) conducting DoS attacks** are a moderate likelihood risk but could cause severe downtime, disrupting critical operations and affecting customer trust. **Hardware or software failure** is also a realistic concern, as disk or system malfunctions could temporarily halt database availability, directly affecting daily business functions.

## Remediation Strategy

To mitigate the risk of **data exfiltration by competitors**, implementing **Public Key Infrastructure (PKI)** and **end-to-end encryption** ensures sensitive information is protected during transmission. Applying the **principle of least privilege** and enforcing **role-based access controls** restricts access to critical data, reducing insider threats. To prevent **DoS attacks by hackers or APTs**, deploying **firewalls, intrusion detection/prevention systems (IDS/IPS), and rate-limiting mechanisms** can block excessive traffic. Finally, to address **hardware or software failures**, implementing **regular system monitoring, redundant storage, and automated backups** ensures business continuity and reduces downtime during critical operations.